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Ex.No. 14: 5G Throughput vs SNR with Modulation Labels

OBJECTIVE 1:

CODE:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP * (1 - exp(-SNR/10));
% Display values
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR, Throughput, '-o', 'LineWidth', 2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
```

% ----- Graph 2 -----

subplot(2,1,2)

bar(Throughput)

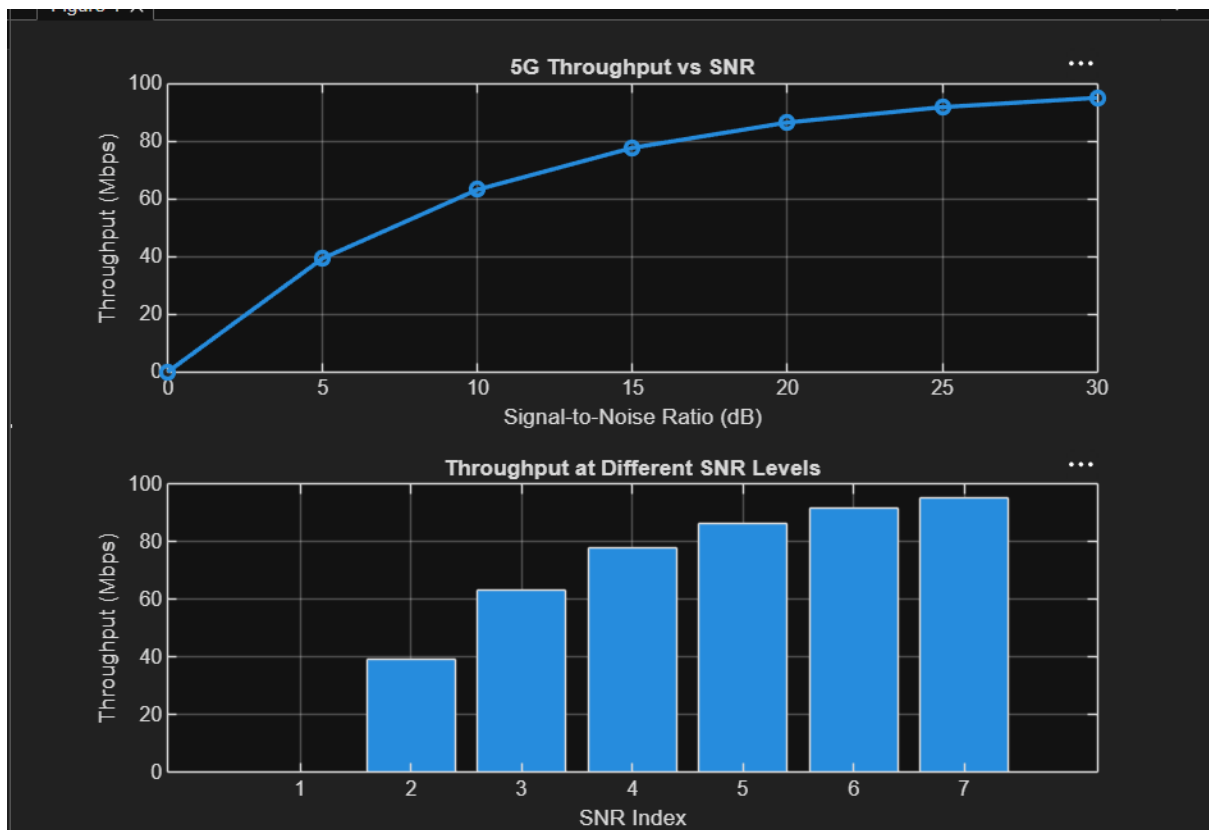
title('Throughput at Different SNR Levels')

xlabel('SNR Index')

ylabel('Throughput (Mbps)')

grid on

O



OBJECTIVE 2:

CODE:

clc;

clear;

```

close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)
Reliability = (1 - BER) * 100;

% Display values
disp('SNR(dB)  BER  Reliability(%)')
disp([SNR' BER' Reliability'])

figure

% ----- Graph 1 -----
subplot(2,1,1)
semilogy(SNR, BER, '-o', 'LineWidth', 2)
title('SNR vs Bit Error Rate in 5G')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Bit Error Rate (BER)')
grid on

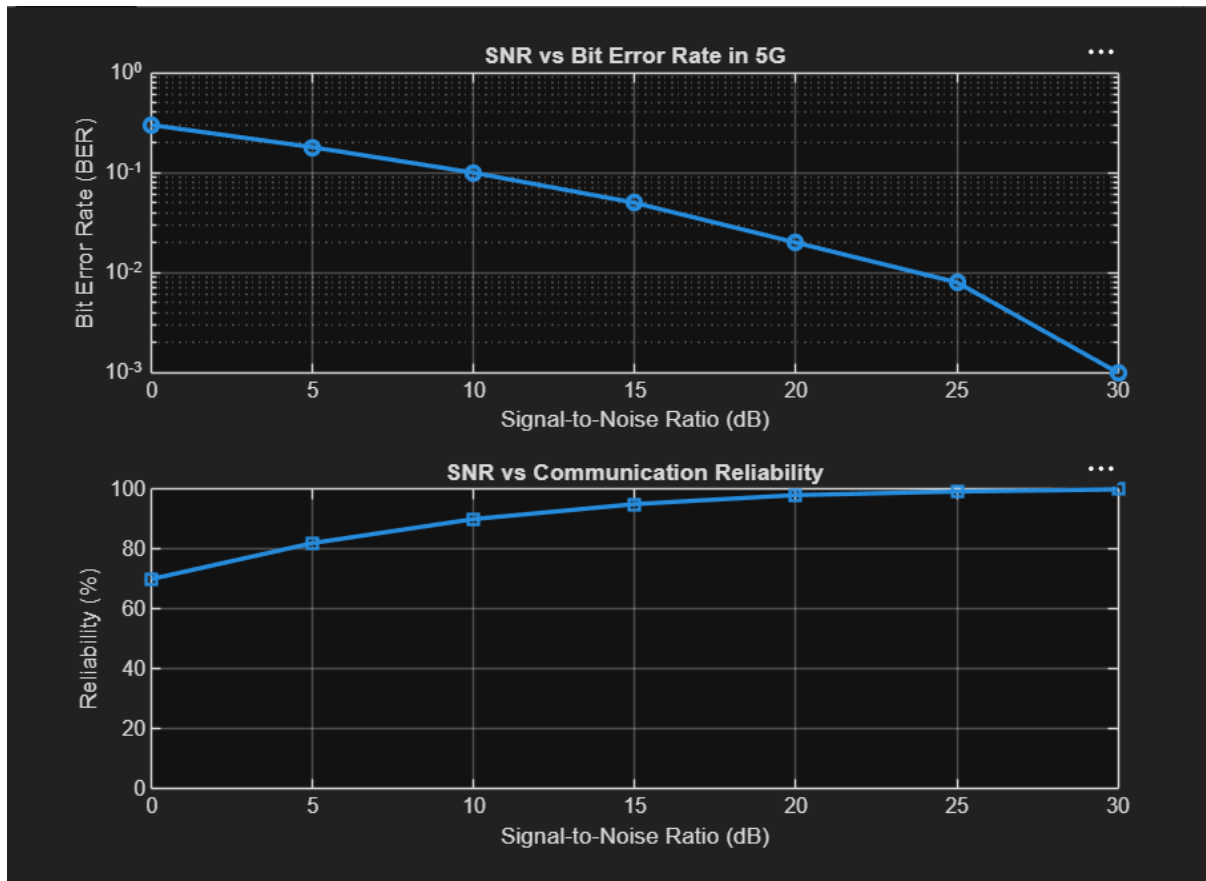
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR, Reliability, '-s', 'LineWidth', 2)
title('SNR vs Communication Reliability')
xlabel('Signal-to-Noise Ratio (dB)')

```

```
ylabel('Reliability (%)')
```

```
grid on
```

```
ylim([0 100])
```



OBJECTIVE3:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Modulation schemes
```

```
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
```

```

% Required SNR (dB)
SNR = [5 10 18 25];
% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
% Display values
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation', SNR', Efficiency', ...
    'VariableNames', {'Modulation','SNR_dB','Efficiency'}))
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(1:4, SNR, '-s', 'LineWidth', 2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on

% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR, Efficiency, '-o', 'LineWidth', 2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')

```

```
ylabel('Spectral Efficiency (bits/s/Hz)')
```

```
grid on
```

